Architectural Particular Specification

PLASTERING, RENDERING AND SCREEDING

SECTION 8.2

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SECTION 8.2

PLASTERING, RENDERING AND SCREEDING

PART 1 GENERAL

1.01 SUMMARY OF WORK

A. Section Includes: The Work of this Section includes, but is not limited to, the requirments of the following:

1. Plastering;
2. Rendering;
3. Screeding

1.02 REFERENCES

(NOT USED)

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

A. General Requirements

1. Plastering and rendering shall be generally to B.S. 5492 : 1990 and B.S. 5262 : 1991 respectively.
2. Floor screed and insitu floor finishes shall be to B.S. 8204 : Part 1 : 1987.
3. Ceramic floor tiles and mosaics, and terrazzo tile and slab, natural stone and composition block floorings shall be to B.S. 5385 : Part 3 : 1989 and B.S. 5385 : Part 5 : 1994 respectively.
4. Internal and external ceramic wall tiling and mosaics shall be to B.S. 5385 : Part 1 : 1995 and B.S. 5385 : Part 2 : 1991 respectively.
5. Wall and ceiling finishes shall be formed to a true plane, correct line and level.
6. All angles and corners shall be formed to right angles unless otherwise specified.
7. All walls and reveals shall be finished to plumb and square.
8. Maximum deviation permitted in finished wall and ceiling surfaces shall be 3mm from a 1800mm straight edge.

1.05 ENVIORNMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

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1.07 SUBMITTALS

(NOT USED)

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

(NOT USED)

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 MATERIALS

A. Cement

1. Ordinary and rapid-hardening Portland cement shall comply with latest requirements of B.S. 12.

2. Where special requirements are shown on the drawings for other types of cement, they shall conform to latest British Standard requirements as follows :-

   a. Rapid Hardening Cement - B.S. 12

   b. Portland Blast furnace Cement - B.S. 146

   c. Low Heat Portland Cement - B.S. 1370

   d. Sulphate Resisting Portland Cement - B.S. 4027

   e. High Slag Portland Blast furnance Cement - B.S. 4246

   These other types of cement shall only be used with the written approval of the Architect.

3. High Alumina Cement shall not be used in any part of the works unless specified elsewhere in this contract.

4. All cement shall be from a manufacturer of good repute and shall be subject to the Architect's approval.

5. Each consignment of cement shall be accompanied by a manufacturer's test certificate showing the quantity, delivery note number, date consigned and the results of recent tests on representative samples.

6. All cement to be used on the works shall be supplied from the same brand and source unless alternate brands or sources are approved by the Architect. Cement of different approved brands shall be separately stacked or stored and shall not be used in the same pour.

7. The Architect may require that any cement, delivered to the site, or elsewhere, for used in the works shall be sampled and tested in accordance with B.S. 4550 : 1978. Any batch of cement so tested which fails to comply with this Specification will be rejected. No deliveries in bulk shall be commenced until samples are approved.

8. All cement to be used on to the site or elsewhere for the works shall be delivered in the original sealed and branded bags of the manufacturer or in bulk containers endorsed by the Architect.

9. Cement, unless delivered in bulk, shall be stored in a weatherproofed shed well protected from the weather and dampness, the floor of which shall be raised at least 150mm above the ground to allow free air circulation. Cement delivered in bulk shall be stored in a weatherproof silo. All cement shall at all times be protected from deterioration or contamination

10. Bagged cement shall not be stacked up more than 15 bags high in order to prevent hard lumps being formed.

11. Any damaged bags or bags in which there are hardened lumps or cakes of cement will be rejected and must be removed from the site by the Contractor.

12. The cement store shall be so arranged that batches of cement can be used in the order in which they were delivered.

13. Any cement which, upon inspection, is considered by the Architect to have been in any way affected by dampness or hydration or otherwise unsuitable for use will be condemned without further tests and shall be removed from the site forthwith.

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B. Sand

1. Sand shall be clean, hard, durable crushed or sea sand obtained form the Hong Kong Government Monopoly, it shall be thoroughly washed and sieved to remove all dirt, shells and others deleterious matter before use.

2. Sand shall be free from any deleterious materials such as clay, flaky or elongated particles, mica, shale or other laminated materials, organic impurities, iron pyrites and shall not contain more than 1% of soluble sulphate salts. In general, and except as specified herein, the sand is to comply with latest B.S. 1199 and 1200.

3. Sand for mixed not incorporating lime shall be crushed granite or other durable rock fines or natural sand free from salt and shall conform to the grading limits set below :-

B.S. Sieve	Percentage By Weight Passing B.S. Sieve
5.00mm	100%
2.36mm	90 - 100%
1.18mm	70 - 100%
0.60mm	40 - 80%
0.30mm	5 - 40%
0.15mm	0 - 10%

4. Sand for mixes incorporating lime shall be clean natural sand free from salt and shall conform to the grading limits set below :-

B.S. Sieve	Percentage By Weight Passing B.S. Sieve
2.36mm	100%
1.18mm	90 - 100%
0.60mm	55 - 100%
0.30mm	5 - 50%
0.15mm	0 - 10%

5. Sand other than the 5.00mm size whose grading falls outside the limit set above by a total amount not exceeding 5% will be accepted.

6. The quantity of clay, fine silt and fine dust present when determined in accordance with latest B.S. 812: Part 105: Section 103.1 or B.S. 812: Part 106 must not exceed 10% by weight.

7. Where screeds, backing to tiles, rendering, plastering or other finishes are to be used externally, the use of sea sand washed or otherwise is strictly prohibited.

C. Lime

1. Lime used for mortar is to be hydrated lime or quicklime which complies with B.S. 890 : 1995 free from lumps and is to be delivered to the site in sealed bags bearing the manufacturer’s name or brand. The use of shell lime for mortar will not be permitted.

D. Lime Putty

1. Lime putty shall be prepared in accordance with B.S. 5492 as follows :-

    a. Add hydrated lime to B.S. 890 to water, mixed to putty with a thick, creamy consistency and left undisturbed for at least 16 hours before use, or

    b. Slaked quicklime. Leave undisturbed for minimum period of 36 hours for grey chalk lime, and 2 weeks for white lime, before use.

E. Additives And Admixtures

1. Plasticisers shall be a proprietary brand complying with B.S. 4887 : Part 1. Submit manufacturer's data sheets and other information as required by the Architect.

F. White or coloured cement

1. White or coloured cement is to be approved brand of similar quality and of the colours required for the work. It is to be delivered to the Site in sealed drums bearing the name of the manufacturer and the brand of cement.

2. Where the term "coloured" is used, the meaning shall also include white colour.

G. Water

1. Water for finishing work shall be clean and uncontaminated potable water from Government main supply or any other approved source free from harmful matters.

2. If water is taken from a source other than Government main supply, it shall be tested in accordance with B.S. 3148 when so required by the Architect.

H. Bonding agent

1. Bonding agent to be compatible with background and finish, designated suitable for internal or external use and to be proprietary brand specified or approved by the Architect.

2. Bonding agent for gypsum plaster to be to B.S. 5270: Part 1.

I. Waterproofing additives

1. Finishing specified to be "waterproofed" shall be treated with an approved proprietary waterproofing additives (if not expressly specified otherwise).

2. All waterproofing additives shall be applied :-

   a. In strict accordance with manufacturer's instructions,

   b. Under the direct supervision of the supplier.

J. Hardener

1. Surface hardener shall be an approved proprietary non-metallic liquid hardener or dust proofer.

2. All hardeners shall be applied :-

   a. In strict accordance with manufacturer's instructions,

   b. Under the direct supervision of the supplier.

K. Metal lathing

1. Metal lathing shall be comply with B.S. 1369: Part 1

   a. Plain expanded type of 6mm short way mesh coated with tight coat galvanising and weighing not less than 1.6 kg/m2.

   b. Ribbed expanded type similarly coated and weighing not less than 2.25 kg/m2.

2. Metal lathing shall be :-

   a. Galvanized or coated with black bitumen-based coating before leaving the manufacturer's workshop, and

   b. Completely free from rust.

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L. Plaster beads

1. Plaster beads shall be approved galvanised steel beads complying with B.S. 6452: Part 1.

2. Plaster beads to be used shall be :-

    a. Corner beads, or

    b. Movement joint beads, or

    c. Plaster stops as specified.

M. Wire netting

1. Wire netting shall be zinc coated hexagonal steel wire netting complying with B.S. 1485 : 1983 (1989).

2. Unless otherwise specified, wire netting shall have mesh size of 50mm and shall be manufactured with 0.9mm diameter (wire designation 19) annealed mild steel wire complying with B.S. 1052.

N. Staples

1. Staples shall be galvanized steel wire staples complying with B.S. 1494, Part 1.

O. Tying wire

1. Tying wire to be 1.25mm annealed steel wire, galvanized to B.S. 443.

P. Gypsum plaster

1. Gypsum plaster shall be of retarded hemihydrate gypsum plaster - Grade B of the following types complying with B.S. 1191: Part 1:-

1.	Undercoat plaster	-	Brown plaster or Metal lathing plaster,
2.	Final coat plaster	-	Type a, Finish plaster or Board finish plaster, Type b.

2. Gypsum plaster shall be delivered in manufacturer's sealed containers and stored in weathertight structures with a raised floor.

Q. Keene's plaster

1. Keene's plaster shall be an approved anhydrous gypsum finishing plaster to B.S. 1191: Part 1, Grade D.

R. Acoustic plaster

1. Acoustic plaster shall be an approved proprietary brand free from asbestos; delivered to site, stored, mixed and applied to manufacturer's instructions by a specialist contractor skilled in the application of the approved materials.

S. Plasterboard

1. Plasterboard shall conform to B.S. 1230: Part 1, "gypsum lath" or "gypsum baseboard" with square edges.

2. Carry plasterboard on edge. Stack plasterboard flat on level surface, off the ground and inside a building. Protect from weather and damp. Programme deliveries to ensure that storage on site is kept to a minimum during periods of high humidity.

3. Nails for plasterboards shall be 2.65mm diameter galvanised steel plasterboard nails jagged shank type to B.S. 1202: Part 1 with length of 30mm for

plasterboard not exceeding 12.7mm thick and 40mm for plasterboard 19mm thick.

4. Reinforcement for joints in plasterboard shall be jute scrim cloth not less than 90mm wide.

T. Cornices & plaster mouldings

1. Cornices and plaster mouldings shall be either :

   a. Formed in cement and sand mortar 1:3 and finished in plaster to an approved profile using steel templates; or

   b. Preformed mouldings by approved manufacturers.

U. Aggregates for in-situ finishings

1. Stone chippings for “Shanghai” plaster are to be clean granite, white stone or marble chippings, graded from 3mm to 5mm, and free from dust or other deleterious matter.

2. Stone aggregate for granolithic finishes are to be natural pebbles, crushed grey granite or white stone to B.S. 882, graded from 3mm to 10mm, and free from dust. Grey granite and natural pebbles shall be clean and free from decomposed or discoloured stone.

3. Marble aggregates for terrazzo finishes are to be angular crushed marble, graded from 3mm to 10mm, clean and free from dust or other contaminants, and of the colour required.

V. Aggregate for lightweight screeds

1. Aggregate for lightweight screeds shall be as follows :-

   a. 5mm exfoliated vermiculite complying with B.S. 3797.

   b. Approved proprietary lightweight beads or granules.

W. Air entraining agent for lightweight screeds

1. Air entraining agent for lightweight screeds shall be an approved admixture which will produce screeds with a density not more than 1,200kg/m2.

X. Vapour barrier

1. Vapour barrier shall be 0.08mm thick polythene sheet.

Y. Non-slip strips

1. Non-slip strips shall be either :-

   a. Carborundum strips composed of one (1) part of cement and one (1) part of carborundum dust, or

   b. An approved proprietary black insert strip.

2. The size shall be 15 x 25mm (width x depth) with slightly curved top.

Z. Dividing strips

1. Dividing strip shall be of brass, aluminium, stainless steel or plastic strip as specified or shown, 3mm thick and depth to the full thickness of finishing.

2. Plastic strip shall be of the colour required.

   a. Generally sides of section to be grooved.

   b. Metal strip - one edge to be cut and bent at 150mm centres, to form lugs.

   c. Plastic strip - to have 5 to 10 mm holes at 150 mm centres with plastic pins inserted to form dowels.

AA. Joint sealant

1. Joint sealant, unless otherwise specified, shall be approved silicone joint sealant.

PART 3 EXECUTION

3.01 PREPARATION

A. Preparation of background generally

1. Hack off concrete projections and fins.
2. Remove efflorescence, laitance, mould oil or other release agents, oil, grease, dirt, dust or other loose materials and contaminants by thoroughly dry brushing or scraping.
3. Protect surfaces from weather and ensure that they are completely compatible with the finish to be applied before starting working.
4. Joints of brickwork, masonry, etc. shall be raked out and brushed to remove loose particles, mortar, etc. to form key for base coat.

B. Preparation of concrete surface for monolithic finish

1. Where finish or screed is to be laid monolithically on concrete base :-
   a. Spray surface with water as soon as concrete has set,
   b. Brush with a stiff broom to remove laitance and loose aggregate to provide a key before concrete has hardened,
   c. Lay finish or screed while concrete is still “green”, - i.e., within 3 hours of casting of concrete.

C. Preparation of aged or hardened concrete surface

1. Where finish or screed cannot be laid monolithically on concrete base, before applying finish or screed :-
   a. Thoroughly hack concrete base to remove laitance and expose coarse aggregate,
   b. Thoroughly clean and well wet the hardened concrete base and remove surplus water,
   c. Brush neat cement slurry into damp surface of the base immediately before applying the finish or screed, or
   d. Apply approved proprietary bonding agent in accordance with manufacturer’s instructions.

3.02 INSTALLATION / APPLICATION

A. Junctions between dissimilar solid backgrounds in the same plane and with the same coating shall be :-

1. provided with a strip of steel lathing extending 150mm (minimum) on either side of the junction,
2. fixed with 40mm nails or with staples to plugs at 100mm centres at each edge.

B. Trueing of surfaces

1. All finishes to walls or ceilings shall be brought to true and even and flat surfaces by means of screeds temporarily fixed at angles and the surfaces trued off with a straight edge before trowelling or floating.

C. Dubbing Out

1. Where applied to uneven surfaces the undercoat to finishings of any type shall be dubbed out as follows :-

   a. dub out in layers of 10mm (maximum) thick in same mix as the first coat.

   b. Allow to dry out before next coat is applied.

   c. Total thickness of dubbing out is not to exceed 25mm.

   d. Cross scratch each coat to provide key.

   The thicknesses shown or given shall be exclusive of any key and any necessary dubbing.

2. For external renders and backing to tiles, the dubbing out of the surface shall be applied as follows :-

   a. For areas not larger than one metre square on plan and not thicker than 8mm at any point, dubbing out or levelling shall be carried out using the same mix as the render to be applied, with or without bonding coat as specified for the render, in a single coat application, trowelled and then scratched to form a key.

   b. Where the area exceeds one metre square or 8mm in depth, reinforcement consisting of 1.5mm by 1.5mm stainless steel welded mesh with wires at 25mm centres shall be shot fixed to the backing concrete. The mesh shall be applied with the horizontal wires outwards and fixed through 30mm diameter stainless steel washers at 300mm centre to centre in each direction. The mesh shall be spaced from the background such that it lies in the centre of the layer of levelling render or dubbing out. The levelling render shall then be applied as described above.

D. Fixing Of Metal Lathing

1. Metal lathing shall be fixed as follows :-

   a. With the largest dimension of mesh running at right angles to direction of supports.

   b. Secured to timber with 30mm galvanized steel staples at 100mm centers, or

   c. Secured to steelwork with 1.25mm diameter galvanized steel tying wire at 100mm centres, or

   d. Secured to concrete or brickwork with nails or staples and plugs at 100mm centres.

2. All cut ends shall be painted with one coat of black bitumen-based coating and turned away from the plastered surface.

3. Metal lathing shall be lapped with a minimum of :-

   a. 25mm at side laps,

   b. 50mm at end laps occurred between supports, and

   c. 25mm at end laps occurred at the supports.

4. All laps shall be secured with 1.25mm diameter galvanized steel tying wires at 75mm centres.

5. Round or square nosed casing beads shall be provided at edges as specified.

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E. Fixing Of Plasterboard

1. Fixing of plasterboard shall comply with B.S. 8212 : 1995

2. Plasterboard shall be fixed to wood or metal bearers as follows :-

   a. Nail or screw boards at each support, at 150mm centres working out from the centre of the board and 15mm (minimum) from edges.

   b. Provide gap of 3 to 5 mm between edges.

F. Joints In Plasterboard

1. Joints, etc. in plasterboard shall be treated as follows :-

   a. Fill joints and internal angles with board finish plaster.

   b. Reinforce board joints with jute cloth scrim pressed and trowelled flat into plaster. Allow plaster to set but not dry out before plastering commences.

G. Fixing Of Plaster Beads

1. Plaster leads shall be fixed by :-

   a. Plaster dabs, or

   b. Galvanized nails or staples.

2. Plaster beads shall be fixed to in such a way that the mesh is as close to the wall as possible to avoid it from spring out of the plaster.

3. Plaster beads shall be fixed to plumb and level and the finishing coat shall be trowelled flush with the bead.

4. Excessive plaster polishing at the arrises should be avoided.

5. Plaster beads shall be butt or mitred jointed at ends.

H. Spatterdash

1. Off-formed concrete surfaces that require plastering shall have a spatterdash coat applied to form key and consisting of :-

   a. One (1) part cement, and

   b. Two (2) parts of coarse sand or granite fines

   c. mixed with sufficient water to a thick slurry.

2. Apply spatterdash to concrete vertical surfaces and soffits immediately after striking of formwork.

3. Clean down surfaces, removing mould oils or other release agents, efflorescence, laitance, etc.

4. Spatterdash shall be thrown on with a hand trowel or scoop to form key to a thickness not exceeding 6mm with a rough cast finish.

5. To prevent rapid loss of moisture and insufficient hydration, spatterdash shall be well wetted one (1) hour after application and left to cure and harden slowly before applying undercoat.

6. Surface of spatterdash shall be lightly brushed to remove any loose particles and well wetted to minimise absorption two (2) hours before application of finishes.

7. The Main Contractor's attention is particularly drawn to that formwork to vertical surfaces of concrete to be covered with tiling or other applied finish shall be stripped within 48 hours of casting with spatterdash applied immediately as specified above. Should the contractor fail to strip the formwork within the period, or if spatterdash coatings are not applied on the concrete surfaces later than four (4) hours after striking of formwork, the concrete surfaces that require

forming key shall be roughened by bush-hammering or abrasive blasting using an approved hand or mechanical tool by the Main Contractor at no extra cost to remove the cement surface skin and expose the coarse aggregates without chipping, cracking or loosening any individual aggregates before applying finishing

I. Plastering And Rendering Generally

1. Surfaces to be plastered shall be well wetted before work is commenced.
2. Base coat shall be :-
    a. Plumb and level,
    b. Laid on evenly,
    c. Straightened with a rule and,
    d. Scratched to form key.
3. Allow seven (7) days for undercoats to dry out thoroughly before applying next coat. Cross scratch undercoats to provide key for next coat.
4. Finishing coat shall be trowelled with a steel trowel to a smooth and even surface with sharp straight corners or round arrises as required.
5. Mixing of plaster and mortars, except for small quantities or otherwise approved in writing by the Architect, shall be carried out with suitable type of mechanical mixer.
6. All mixes are to be measured by volume unless otherwise specified.

J. Internal Lime Plaster On Solid Background

1. Internal lime plaster shall be applied on solid backgrounds in two (2) coats as follows:
    a. Base coat, 10mm (maximum) thick to walls and 5mm (maximum) thick to soffits, consisting of one (1) part cement, four (4) parts lime putty and, sixteen (16) parts sand.
    b. Finishing coat consisting of either one (1) part cement, twelve (12) parts lime putty and, thirty (30) parts sand, or finishing coat of gypsum plaster added with 25% of lime putty.
2. Total thickness of internal lime plaster shall not exceed 15mm to walls and 10mm to soffits.

K. Internal Lime Paster On Metal Lathing

1. Internal lime plaster on metal lathing shall be applied in three (3) coats as follows:-
    a. First and second coats consisting of one (1) part cement, two (2) parts lime putty and six (6) parts sand.
    b. Finishing coat consisting of cement, lime and sand 5mm thick - mix 1 : 12 : 30 with smooth finish, or finishing coat of gypsum plaster with 25% of lime putty added with smooth finish.
2. Total thickness of plaster shall not exceed 13mm measured from the outer face of the lathing.

L. Angles

1. Arrises shall be square or pencil rounded, as required.
2. Provide and fix metal corner beads, plaster stops and movement joints when specified including nailing, stapling or fixing with plaster dabs and trowel the finishing coat flush with the bead.

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3. Coved or moulded cornices shall be either :-

    a. Formed with a backing of cement and sand (1:3) with finishing coat of same plaster used for adjacent surfaces, finished with a steel template to a smooth finish, or

    b. Preformed cornices from an approved manufacturer fixed in accordance with manufacturer's recommendations.

M. Skim Coat Plaster

1. When skim coat plaster is specified, the base coat can be saved if the surfaces of backgrounds for plastering are true and level. The skim coat shall be 7mm thick throughout and composed of :-

    a. One (1) part cement,

    b. Two (2) parts lime putty and,

    c. Six (6) parts sand.

N. Gypsum Plaster On Solid Background

1. Gypsum plaster on concrete, concrete blocks, bricks and similar solid surfaces shall be applied generally in two coats with total thickness not exceeding 10mm and as follows :-

    a. First coat consisting of Browning plaster and sand 1:2.

    b. Finishing coat of neat finish plaster, or finish plaster with up to 25% of lime putty added, trowelled to a smooth finish.

O. Gypsum Plaster On Plaster Board

1. All joints between plasterboards and angles between walls and ceilings shall be filled with final coat gypsum plaster as specified in A5.2.16-a2.
2. Press 90mm wide strips of jute scrim cloth joint reinforcement into the plaster, trowel flat and allow the plaster to set.
3. All joints shall be scrimmed before plastering.
4. Do not wet plasterboard surfaces before plastering.
5. A thin coat of neat gypsum plaster shall be applied over the whole surfaces of plasterboards to level up.
6. Gypsum plaster as a finishing coat 5mm (maximum) thick shall be applied on plasterboards trowelled to a smooth surface using as little water as possible.

P. Gypsum Plaster On Metal Lathing

1. Gypsum plaster on metal lathing shall be applied in three (3) coats with the total thickness not exceeding 13mm measured from the outer face of the lathing.
2. First coat shall consist of metal lathing plaster and sand 1 : 1½. Second coat shall consist of browning plaster and sand 1 : 2. The total thickness of first and second coats shall not be more than 10mm measured from the outer face of the lathing.
3. Finishing coat, 3mm (maximum) thickness shall be of neat finish plaster, or finish plaster with up to 25% of lime putty added, and finished smooth.
4. Each successive coat shall be allowed to dry thoroughly before application of the further coat.

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Q. Trowelled Concrete Finish

1. Where trowelled concrete is specified, the surface of the concrete is to be trowelled smooth and flat using a steel hand or power trowel, or float, leaving the surface free from trowel or float marks.

R. Cement And Sand Paving

1. Cement and sand paving shall consist of :-

   a. One (1) part cement, and

   b. Three (3) parts sand.

2. Coloured cement and sand paving shall be laid in 2 coats using Ordinary Portland cement for first coat and an approved coloured cement for the finishing coat.

3. Cement and sand paving shall be trowelled smooth with a steel trowel or power float to a smooth surface free from blemishes, or floated with a wooden float to give an even overall surface, unless otherwise specified.

4. Cement and sand paving shall be laid in alternate bays of 15 square metres (maximum) with length not more than 1½ times of the width. A minimum of 24 hours between laying adjoining bays shall be allowed.

S. Cement and Sand Paving With Hardener

1. Hardener shall be applied in strict accordance with manufacturer’s instructions

2. Paving with hardener shall be laid while the concrete surface is still green, otherwise bonding agent compatible to the hardener shall be provided at the Main Contractor’s own expenses.

3. Paving with hardener shall be laid in panels of size as recommended by the manufacturer of the floor hardener.

T. Surface Finishes For Screeds & Paving

1. Surface finishes on flooring shall be applied as soon as the compaction is completed.

2. Surface finishes shall be applied in such a way that no excessive laitance is being brought to the surface ad remove any which appears.

3. No wetting of surface is allowed during applying surface finishes.

4. Smooth or trowelled surface shall be finished with a steel trowel or power float to give a smooth untextured surface, free from blemishes.

5. Wood floated surface shall be finished with a dry wood float to give an even overall surface.

6. Combed or brushed surface shall be finished with a comb of suitable size and type or a stiff bristled brush to give a suitable size and type or a stiff bristled brush to give a slightly roughened even texture whilst the surface is still green.

7. Ribbed surface shall be finished with the edge of a board with suitable length and tamped lightly on the surface to form regular, parallel and equally spaced shallow ribs perpendicular to the direction of traffic flow as necessary.

8. Textured surface shall be finished by stippling, scrapping or other means to produce an approved textured finish.

9. Rough cast surface shall be finished by throwing on to the undercoat a wet mix of aggregate and cementitious material.

10. Machine applied textured finish shall be applied in strict accordance with the manufacturer’s instructions.

U. Finished Thickness Of Cement And Sand Paving

1. Unless otherwise specified, the finished thickness for cement and sand paving shall be as follows :-

   a. 40mm (normal) thick for floor and treads.

   b. 20mm (normal) thick for risers.

2. The paving shall be laid level, to falls or to falls and currents as required.

V. Curing Of Floor Screeds And Pavings

1. On completion of the surface finish to floor screeds and pavings they are to be covered with heavy duty polythene sheet or tarpaulins, well lapped at edges and left to set for not less than seven (7) days.

W. Floor And Wall Screeds

1. Floor and wall screeds shall consist of :-

   a. One (1) part cement and,

   b. Three (3) parts sand.

   c. Granite fines shall be used instead of sand, when required, to avoid efflorescence on the surface of the finish. The minimum water consistent with workability shall be used.

2. Floor and wall screeds shall be :-

   a. Floated with a wooden float to obtain the required thickness,

   b. Floated to true even surfaces and,

   c. Finished to true levels, to falls or to falls and currents as shown on drawings.

3. Monolithic and bonded screeds shall be laid in one coat. Screeds shall be laid & compacted level or to falls and currents, as required.

4. Before laying, the Main Contractor shall prepare :-

   a. All level and reference lines,

   b. Wooden/metal level marks at 1,500mm intervals at both directions.

5. After the screed is laid, the surface shall be :-

   a. Trowelled with a teak level rod, minimum 1,800mm long by 50mm wide with continuous surface and,

   b. Finally finished with a wooden float.

6. For screed over 40mm thick, mix to be one (1) part cement, 1½ part sand or granite fines and three (3) parts coarse aggregate graded 10mm down with at least 75% being retained on a 5mm B.S. sieve.

7. The surface of screed for thin flexible tile or sheet shall be finished to a smooth surface with a steel trowel or power float and free from blemishes.

8. Floor and roof screeds shall be laid in alternate bays of 15 square meters (maximum) with length not more than 1½ times of the width. A minimum of 24 hours between laying adjoining bays shall be allowed.

X. Thickness Of Floor And Wall Screeds

1. Thickness of floor screeds shall be as follows :-

1.	Laid monolithically with concrete base	-	15mm (minimum) thick
2.	Bonded to a hardened concrete base	-	20mm (minimum) thick
3.	Not bonded to the bases	-	50mm (minimum) thick including tile finish
4.	Floating	-	65mm (minimum) thick including the finish

2. Where different tile finished are specified at adjacent areas, the thickness of floor screeding shall be varied with the relative tile thickness to bring all adjacent finishes to the same level. However, unless otherwise specified, the finished floor thickness including tile finish shall not be less than 40mm.

3. Thickness of wall screed shall be not less than 10mm with surface lightly scratched to form key.

4. Subject to loading capacity of the existing structural slabs, thickness of internal floor screed shall be not more than 60mm.

Y. Lightweight Screeds

1. Lightweight aggregate screeds shall consist of cement and lightweight aggregate 1:8 for roofs and 1:6 for floors unless contrary to manufacturer's recommendations.

2. Lightweight screed may be formed by air entraining agent which shall have a dry density of not more than 1200 kg/m3. Add additive to the mix in the proportions recommended by the manufacturer.

3. Lightweight screeds shall be laid to a minimum thickness of 50mm (excluding topping).

4. Lightweight screeds shall be finished with cement and sand or granite fines 1:4 topping 15mm (minimum) thick laid monolithically with the screed.

5. Vapour barrier shall be laid under lightweight roof screeds and lapped 150mm at joints

Z. Filling To Levels

1. When structural slab sinkings are required to be levelled up, they shall be filled with screed consisting of :-

   a. One (1) part cement and,

   b. Four (4) parts sand,

   c. with such falls as necessary to receive the required floor finish.

2. When the required thickness of the filling exceeds 20mm, unless otherwise specified, the remaining depth shall be filled up with mixture of :-

   a. One (1) part cement,

   b. Three (3) parts sand and,

   c. Six (6) parts approved light-weight aggregates.

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AA. Pipes Through Roofs

1. Where pipes of less than 50mm diameter pass through flat roofs, 150 x 150 mm surround shall be formed with cement and sand or granite fines 1:3 around pipe sleeve, projecting 150mm above finished roof level.
2. Top of surrounds shall be finished to slope to shed water away from pipe.
3. For groups of pipes in close proximity, combined surrounds shall be formed.

BB. Cement And Sand Rendering

1. Cement and sand rendering shall consist of one (1) part Portland cement and three (3) parts sand or granite fines, to be applied as follows :-
    a. in one coat not exceeding 10mm thick, or
    b. in two coat exceeding 10mm thick with the finishing coat 5mm thick.
2. Render shall be float finished or trowelled to the surface finish specified.
3. Cement rendering for unexposed surfaces shall be as follows :
    a. Rough rendering consisting of one (1) part cement and, four (4) parts sand and,
    b. Floated to the thickness of the grounds or as otherwise required to fill in voids behind skirtings, dubbing out to uneven surfaces, etc.

CC. Waterproof Cement Rendering

1. All waterproof admixtures and/or additives shall be used, mixed, applied and cured strictly in accordance with the manufacturer's instructions.
2. Unless otherwise specified, waterproof admixtures or additives shall be applied to the following areas :-
    a. Floor and skirting screeds in toilets and lavatories, janitor's rooms, plant rooms, etc.
    b. Floor and wall screeds to exterior balconies and open staircases,
    c. Rendering to roof water tanks internally and externally,
    d. Rendering to top and sides of copings, parapets and boundary walls as well as covers of pipe duct heads and,
    e. Rough rendering to internal wall surfaces at backs and sides of built-in closets.

DD. Cement / Lime External Rendering

1. Cement / lime external rendering shall be applied in two (2) coats as follows :-
    a. Base coat, 10mm (minimum) thick, consisting of one (1) part cement, two (2) parts lime putty and, six (6) parts sand.
    b. Finishing coat, 10mm (maximum) thick, consisting of one (1) part cement, three (3) parts lime putty and, six (6) parts sand.
2. Total thickness of external rendering shall not exceed 20mm.

EE. Exposed Aggregate Rendering Or "Shanghai" Plaster

1. Exposed aggregate rendering or "Shanghai" plaster shall be 20mm thick, applied in two (2) coats as follows :-

   a. Base coat 10mm thick, consisting of cement and sand 1:3.

   b. Finishing coat 10mm thick, consisting of cement and stone chippings 1:1, with the chippings mixed in one of the proportions measured by volume, as shown below :-

		STONE CHIPPINGS		MARBLE CHIPPINGS		
COLOUR	CEMENT	DARK GREY	LIGHT GREY	WHITE	BLACK	WHITE
Dark	Ordinary	70%	---	20%	10%	---
Medium	White	60%	20%	20%	---	---
Light	White	5%	75%	---	---	20%
White	White	---	---	---	---	100%

2. The finishing coat is to be well washed and scrubbed just prior to initial set on order to remove surperfluous cement and expose the aggregate.

FF. Coloured Finish To Wall

1. Coloured cement and sand or granite fines and coloured granolithic finish shall be laid in two (2) coats as follows :-

   a. First coat consisting of cement and sand or granite fines 1:3.

   b. Finishing coat as A5.3.33 or A5.3.36 and 5mm thick (minimum) for cement and sand or granite fines finish and 10mm thick (minimum) for granolithic finish using coloured cement. The second coat shall be applied before the first coat has set.

GG. Bedding Of Non-Slip Strips

1. Carborundum strip shall be bedded into groove and finished to project 3mm above finished surface with a rounded profile, and are to be set 40mm back from the edge of nosings.

2. Proprietary non-slip strips shall be bedded or fixed in accordance with the manufacturer's instructions.

HH. Slurry

1. Slurry shall be cement and water mixed to a creamy consistency.

2. Cement for slurry shall be :-

   a. White cement for white or light colour tiles,

   b. Coloured cement for coloured tiles.

II. Sealing Joints

1. Joint sealant for sealing joints shall be used and applied strictly in accordance with the manufacturer's instructions.

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3.03 CLEANING

(NOT USED)

3.04 PROTECTION

A. As soon as it is sufficiently hardened, protect all surfaces from damage by covering with :-

1. A 50mm layer of damp sand, saw dust, or
2. Canvas, or
3. Waterproof sheeting, mats, or
4. Other suitable materials.
5. Keep covered for 7 days (minimum).

B. Prevent excessively rapid or localized drying out of wall and ceiling finishes by approved means.

C. Protect surface of floor screed or in-situ finishes from wind and sunlight immediately after laying.

D. All works surfaces shall be protected from discolouration or physical damage until completion.

E. Where there is heavy traffic during construction period, the Main Contractor shall provide temporary plywood covering at not extra cost for protection of the finished tile surfaces to the satisfaction of the Architect.

F. All finished works shall be cleaned and free from cement slurry upon completion of the work.

END OF SECTION